Instructions:

1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

a[0]=

```
1   #include<stdio.h>
2   void func1(int *arr){
3     for(int i=1; i<=5;i++)
4       (i-1)[arr] = (i%5);
5   }
6   void func2 (int a[1] , int arr){
7     a[0] = arr + 3;
8   }
9  int main(){
10    int arr[6] = {2,2,2,2,2};
11    func1(arr+1);
12    for(int i=0; i<6; i++)
13          printf("%d ", i[arr] );
14    printf("\n");
15    func2(arr, *arr);
16    for(int i=0; i<6; i++)
17          printf("%d ", i[arr] );
18    return 0;
19 }
```

a[0]= [redacted]

arr[0] = 1 , arr[1] = 2 , arr[2] = 3 , arr[3] = 4, arr[4] = 0,

a[0]=3

[2, 1, 2, 3, 4, 0]

0 1 2 3 4

a[0] = 3

Output
2 1 2 3 4 0
[redacted] 2 1 2 6 7 3

Will the answer change if we write func2 as:

```
void func2 (int a[1] , int arr){
  int temp = arr + 4;
  a = &temp;
}
```

Explain your answer

No, as temp stores address of (arr+4) element.

Q. 2: Write the output of the following sequence of instructions **long arr2[90]; int arr1[80], a1=(&arr2[60]), a2=(&arr2[10]), b1=(&arr1[60]), b2=(&arr1[10]); printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

~~a=59~~ a

Ans: 0 0

Because $a_1 = (\&\, arr2[60])$, here, a_1 stores address of 1st box of arr2 array and then 60 tells to move 60 places, to index 59 & then change it's value accordingly.

$a_1 = (\&arr2[60])$ doesnot stores address of 60th element of array arr2.